

IM POLINY OSIPENKO, Russian Federation

WMO: 314160

Lat: 52.4207N

Lon: 136.4799E

Elev: 231

StdP: 14.57

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.8	-35.1	-44.0	0.4	-39.2	-39.6	0.6	-34.8	21.7	12.2	16.7	11.2	0.5	0	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	85.3	69.9	81.9	68.1	78.6	66.3	72.5	82.1	70.2	78.9	68.1	76.0	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.2	108.3	77.9	67.1	100.4	75.5	65.0	93.3	73.3	36.5	81.9	34.4	79.0	32.7	76.3	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	12.8	10.9	-44.5	90.9	5.1	3.6	-48.2	93.5	-51.1	95.6	-54.0	97.6	-57.6	100.2		
(5)			-44.4	75.8	4.8	2.9	-47.9	77.9	-50.7	79.5	-53.4	81.2	-56.9	83.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.7	-13.1	-4.6	13.3	33.5	47.9	59.7	66.0	63.9	52.8	35.5	10.0	-10.6		
	DBStd	29.76	11.61	10.11	10.03	8.20	8.33	8.21	6.95	5.86	6.89	7.70	12.87	11.94		
	HDD50	8849	1955	1528	1137	498	143	12	0	0	48	451	1199	1878		
	HDD65	13087	2420	1948	1602	945	533	200	72	93	368	914	1649	2343		
	CDD50	1445	0	0	0	3	77	303	497	431	132	2	0	0		
	CDD65	210	0	0	0	0	3	42	104	58	3	0	0	0		
	CDH74	1965	0	0	0	1	68	481	948	442	25	0	0	0		
	CDH80	515	0	0	0	0	16	122	298	79	0	0	0	0		
	WSAvg	3.9	2.5	3.1	4.4	5.1	5.5	4.3	3.9	3.6	3.4	4.0	3.5	2.9		
	PrecAvg	19.2	0.4	0.3	0.5	1.1	1.9	2.3	3.3	3.5	2.6	1.8	0.9	0.6		
	PrecMax	26.8	2.1	1.3	1.6	3.3	4.7	5.7	8.3	9.3	5.7	5.6	2.3	2.4		
	PrecMin	9.8	0.0	0.0	0.0	0.2	0.2	0.4	0.2	1.1	0.2	0.0	0.0	0.0		
	PrecStd	3.8	0.4	0.3	0.4	0.7	1.0	1.2	1.9	1.9	1.4	1.2	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	25.5	43.1	66.3	82.0	88.1	91.6	86.0	77.3	62.6	39.5	28.2		
		MCWB	21.7	22.6	35.1	51.5	60.1	69.5	74.7	71.9	64.3	51.6	33.4	26.7		
	2%	DB	16.6	19.7	37.6	58.7	74.3	83.4	86.7	82.1	72.5	55.3	34.3	20.6		
		MCWB	15.3	17.2	30.8	46.0	56.7	66.7	71.4	69.7	61.5	46.0	30.7	19.1		
	5%	DB	11.3	15.6	33.2	53.1	69.8	79.6	83.0	79.0	68.7	51.0	31.4	13.9		
		MCWB	10.2	13.5	27.9	42.4	54.0	64.6	69.8	67.8	58.8	43.4	28.8	12.6		
	10%	DB	5.7	11.8	29.3	47.6	64.9	75.5	79.4	75.7	65.0	47.2	28.4	7.9		
		MCWB	4.7	10.1	25.0	38.4	51.7	62.3	67.9	65.7	56.7	41.1	26.1	6.8		
	0.4%	WB	21.8	22.8	35.6	53.1	62.1	73.0	77.4	74.1	65.7	52.3	34.5	27.2		
		MCDB	23.0	24.8	42.4	64.3	78.3	84.8	86.6	82.6	75.1	60.5	37.8	28.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.6	17.3	31.2	47.0	58.3	68.9	73.4	71.6	62.9	47.8	31.4	19.2		
		MCDB	16.7	19.5	37.0	58.0	72.4	80.1	83.9	79.8	70.5	53.4	33.7	20.6		
	5%	WB	10.2	13.5	28.1	42.6	55.4	66.2	71.2	69.3	60.5	44.4	29.0	12.8		
		MCDB	11.2	15.5	32.7	52.8	67.7	76.3	80.4	76.4	66.5	49.3	31.2	14.0		
	10%	WB	4.7	10.1	25.3	38.9	52.3	63.8	69.1	67.4	58.0	41.7	26.0	6.8		
		MCDB	5.7	11.8	29.2	47.1	63.2	73.3	77.5	73.8	63.5	46.6	28.3	7.9		
	5% DB	MDBR	21.1	24.5	23.6	18.9	21.3	21.5	18.7	17.5	19.6	16.7	17.2	19.1		
		MCDBR	15.7	23.1	25.7	27.7	31.7	27.9	25.2	23.3	25.2	23.1	16.0	15.2		
		MCWBR	14.7	20.7	19.5	16.8	16.2	12.7	11.7	12.0	14.0	14.2	12.6	14.4		
		MCWBR	15.3	21.9	24.1	27.1	30.2	25.1	22.7	20.1	22.0	20.4	14.5	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.293	0.361	0.473	0.550	0.502	0.507	0.428	0.374	0.376	0.309	0.263		
		taud	2.118	2.069	1.919	1.798	1.791	2.014	2.043	2.285	2.367	2.155	2.097	2.077		
	Ebn at Noon		228	254	254	238	226	241	237	253	254	219	200	200		
		Edh at Noon	23	33	47	61	65	52	50	37	30	30	23	20		
(44) All-Sky Solar Radiation	RadAvg		430	806	1292	1614	1612	1794	1607	1373	1084	676	417	324		
	RadStd		21	30	71	99	164	169	178	145	92	55	33	25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	-4.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page